

TS088QDM-LL0 V6.0 Product Specification Rev.0

BUYER	Lenovo
SUPPLIER	ORDOS YUANSHENG Optoelectronics Technology CO., LTD
FG-Code	TS088QDM-LL0

ITEM	BUYER SIGNATURE	DATE
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Prepared	_____	_____
Reviewed	_____	_____
Approved	_____	_____

ORDOS YUANSHENG Optoelectronics Technology CO., LTD

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REVISION HISTORY				
REV.	ECN No.	DESCRIPTION OF CHANGES	DATE	PREPARED
P0		Initial Release	2024-7-19	

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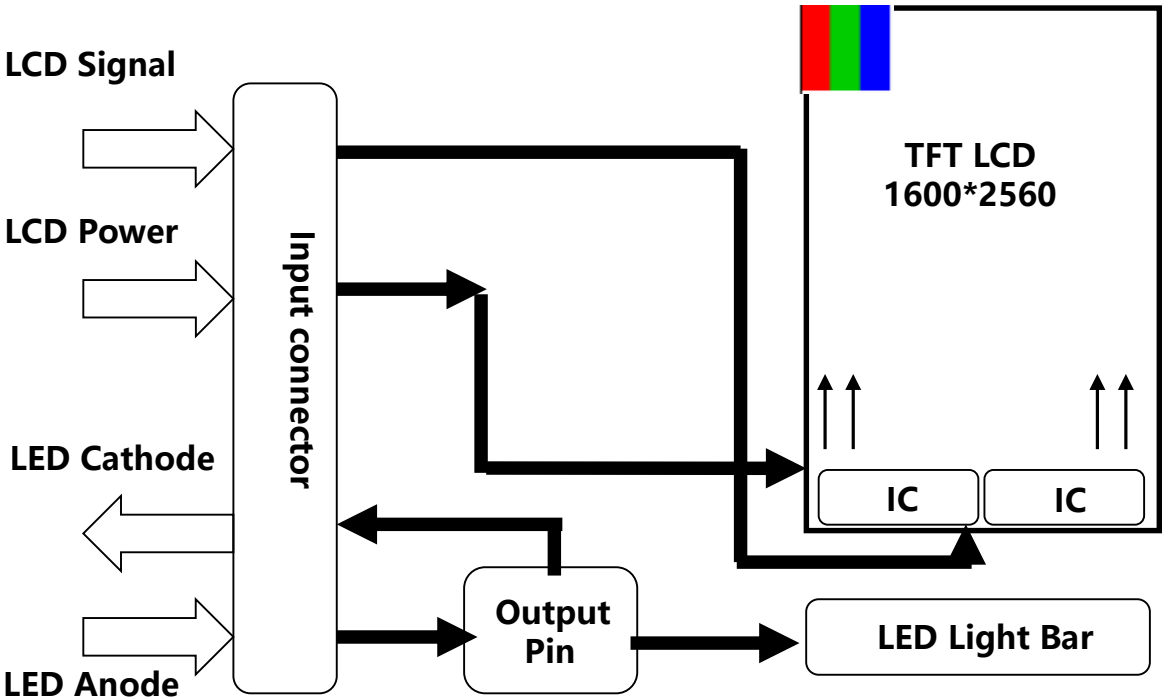
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1.0 GENERAL DESCRIPTION

1.1 Introduction

TS088QDM-LL0 is a color active matrix LTPS LCD Panel using Low Temperature Poly-silicon TFT 's (Thin Film Transistors) as an active switching devices. The LTPS-LCD has a 8.8 inch diagonally measured active area with WQXGA resolutions (1600 horizontal by 2560 vertical pixel arrays). Each pixel is divided into RED, GREEN, BLUE dots which are arranged in vertical stripe and this module can display 16.7M colors.



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1.2 Application

- Tablet PC

1.3 General Specification

The followings are general specifications at the TS088QDM-LL0

<Table 1. LCD Module Specifications>

Parameter	Specification	Unit	Remarks
Active Area	118.464 (H) x 189.5424 (V)	mm	
Number Of Pixels	1600(H) × 2560(V)	pixels	
Pixel Pitch	0.02468(H)×RGB×0.07404(V)	mm	
Pixel Arrangement	Pixels RGB stripe arrangement		
Display Mode	IPS		
Display Colors	16.7M	colors	
Surface Treatment	HC		
Contrast Ratio	1340:1(typ.)1000:(min)		
Viewing Angle(CR>10)	Min. 80°, Typ. 85°	deg.	
Response Time	Max 25	ms	GTG
Color Gamut	DCI-P3 Typ.98%, Min.92% NTSC Typ.96%, Min.91%		
Brightness	Typ.750 Min.640@23mA	cd/m2	@Center
Brightness Uniformity	13point:min:70%; typ:75%; 161point:min:70%; typ:75% ΔUV<0.012		
Power Consumption	LCD: 0.43WTyp(Without TP) BLU: 2..4012W(Max.)	watt	W/O LED Driver; 12 V@500nits ;Mosaic Pattern@60Hz
Outline Dimension	125.96*205.04*2.124@TLCM	mm	
Weight	98g Max (不含保护膜)	gram	

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<Table 2. Touch Panel Specifications>

Parameter		Specification				Remarks
TP Structure		TDDI				
Sensing Method		Self Capacitance				
Number Of Touch		32*50				
Performance	Sensitivity(mm)	Φ4				
	Report Rate	165Hz				
	Finger Separation	≤12mm				
	Response Time	≤30ms				Power up to Active model
	Accuracy(mm)	Center	<=1.0	Edge	<=1.5	@Φ7mm
	Precision(mm)		<=1.0		<=1.5	
	Jitter (mm)		<=0.5		<=0.5	
	Linearity(mm)		<=1.0		<=1.5	
SNR(dB)		45				@Φ4mm
Pen Type & Pen Size		-				
Glove Touch		-				
Hover Reading Height (center)		NA				
Anti Water	Spray	TP works well under below condition, 1,Water-drop(Size ϕ1mm) on the top of TP. 2,Spray on the top of TP.				
	Water Drop					
Power Consumption(Max.) Active/Idle/Sleep		TBD				

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2.0 ABSOLUTE MAXIMUM RATINGS

The followings are maximum values which, if exceed, may cause faulty operation or damage to the unit.

< Table 3. Absolute Maximum Ratings>

Parameter		Symbol	Min.	Max.	Unit	Remarks
Power Supply	LCD Module	VSP	-0.3	6.3	V	Ta = 25 °C Note 1&2
		VSN	-6.3	0	V	
		VDD	/	/	V	
		IOVCC	-0.3	2.10	V	
	Current of 3 Power @White 165Hz with touch	IOVCC	119	152	mA	
		VSP	18	24	mA	
		VSN	13	17	mA	
		Power	416	532	mW	
	Current of 3 Power @White 144Hz with touch	IOVCC	118	151	mA	
		VSP	18	23	mA	
		VSN	10	14	mA	
		Power	397	508	mW	
	Current of 3 Power @White 120Hz with touch	IOVCC	117	150	mA	
		VSP	16	22	mA	
		VSN	9	12	mA	
		Power	382	488	mW	
	Current of 3 Power @White 90Hz with touch	IOVCC	99	128	mA	
		VSP	14	19	mA	
		VSN	10	13	mA	
		Power	337	431	mW	

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< Table 3. Absolute Maximum Ratings>

Parameter		Symbol	Min.	Max.	Unit	Remarks
Power Supply	Current of 3 Power @White 60Hz with touch	IOVCC	90	115	mA	Ta = 25 °C Note 1&2
		VSP	15	19	mA	
		VSN	8	11	mA	
		Power	310	397	mW	
	Current of 3 Power @White 30Hz with touch	IOVCC	76	98	mA	
		VSP	14	18	mA	
		VSN	7	10	mA	
		Power	274	351	mW	
	BLU	VLED	-	26	V	
		ILED	92	-	mA	

- Note:
- These range above is maximum value not the actual operating temperature . Actual Operating temperature is no more than 45°C and temperature refers to the LCM surface temperature ; Length of operation: No more than 8 hours per day, and no more than 4 hours of continuous use one time.
 - BOE is not responsible for product problems beyond the use conditions.

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3.0 ELECTRICAL SPECIFICATIONS

3.1 TFT LCD Module

< Table 4. LCD Module Electrical specifications >

Item		Symbo l	Unit	Min.	Typ.	Max.	Remark
Input power supply		VSP	V	5.7	5.8	5.9	
		VSN	V	-5.9	-5.8	-5.7	
		VDDD	V	-	-	-	
		IOVCC	V	1.90	1.93	2.00	
Current Spec		I _{IOVCC}	mA	-	140	220	单颗IC=3.5mA
		Sleep I _{l ovcc}	mA	-	-	7	
Frame		Hz		-	165	-	-
Input Signal Voltage	H Level	V _{IH}	mV	740	-	-	LP MODE
	L Level	V _{IL}	mV	0	-	550	
Output sign al Voltage	H Level	V _{OH}	V	0.95	1.2	1.3	
	L Level	V _{OL}	mV	-50	-	50	

Note 1: The operation is guaranteed under the recommended operating conditions only. The operation is not guaranteed if a quick voltage change occurs during operation. To prevent noise, a bypass capacitor must be inserted into the line close to power pin. Please make sure all the design settings are used within this range before the panel is initialed. If IOVCC is not typical value, The Panel Character will not be the best.

Note 2. Please make sure that DC is not supplied to LCD for long period. And do not supply voltage to LCD while within "sleep mode".

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3.2 Back-Light Unit

Table 5. LED Driver Electrical Specifications > [Ta =25±2 °C]

Parameter	Symbol	Values			Unit	Notes
		Min.	Typ.	Max.		
LED Supply Voltage	VLED	-	-	26.1	V	Note 1
	VRP	-	-	-	mV	Ripple
LED Forward Current	ILED	-	92	-	mA	4*9
Power Consumption	PLED	-	2.4012		W	
LED Quantity	QLED	-	36	-	EA	
LED Life Time	TLED		15000	-	Hrs	Note 2

- Notes: 1. PLED = VLED ×ILED (Without LED converter transfer efficiency)
2. The life time of LED, 15,000Hrs, is determined as the time at which luminance of the LED is 50% compared to that of initial value at the typical LED current on condition of continuous operating at 25 ± 2°C.



深圳市穗晶光电股份有限公司

SHENZHEN SHUANGGUANG OPTOELECTRONICS CO., LTD.

测试条件：

环境温度：室温，25℃±3℃。

环境湿度：50%±10%。

测试机台：IS机。

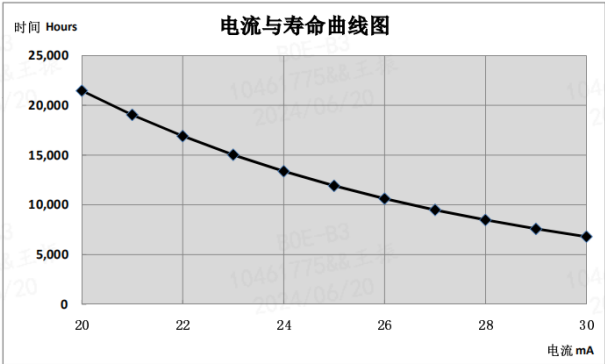
测试方式：恒流测试。

测试电流：20mA~30mA，1mA一级。

寿命标准：衰减到50%

我司SJ-AS26SGLB490D产品在上述测试条件下，不同测试电流与LED 寿命的对应关系如下图所示：

电流（mA）	20	21	22	23	24	25	26	27	28	29	30
寿命(Hours)	21473	19042	16907	15028	13372	11913	10624	9486	8478	7585	6794



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3.3 INPUT TERMINAL PIN ASSIGNMENT

This LCD employs one interface connections, a 61 pin ZIF connector is used for the LCD module electronics interface.

3.3.1 Pin assignment for LCD module

Connector : FH35C-61S-0.3SHW(50) (HRS)

< Table7. Pin Assignment for LCD Module Connector > LCD CN1

Pin No.	Symbol	Description	I/O
1	GND	GND	P
2	C2_S	MIPI input	I
3	B2_S	MIPI input	I
4	A2_S	MIPI input	I
5	GND	GND	P
6	C1_S	MIPI input	I
7	B1_S	MIPI input	I
8	A1_S	MIPI I/O	I
9	GND	GND	P
10	C0_S	MIPI I/O	I
11	B0_S	MIPI input	I
12	A0_S	MIPI input	I
13	GND	GND	P
14	GND	GND	P
15	LED_A	LED+	P
16	LED_A	LED+	P
17	DUMMY	/	/
18	DUMMY	/	/
19	LED_K1	LED-	P
20	LED_K2	LED-	P
21	LED_K3	LED-	P
22	LED_K4	LED-	P
23	DUMMY	/	/
24	DUMMY	/	/
25	RESET	This signal will reset the display screen	I
26	TP_INT	Touch interrupt pin	O
27	TP_SPI_SCK	Flash SPI interface: clock input	I
28	TP_SPI_MISO	Flash SPI interface: data output	O

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< Table7. Pin Assignment for LCD Module Connector > LCD CN1

Pin No.	Symbol	Description	I/O
29	TP_SPI_CS	Flash SPI interface: chip select	I
30	TP_SPI_MOSI	Flash SPI interface: data iutput	I
31	PWM	Output pin for PWM signal of LED driving.	I
32	FTE_Slave	For salve IC detection	I/O
33	FTE	Frame head pulse signal; TE signal and ESD detection;	I/O
34	ID1	0V低电平	O
35	ID0	1.8V高电平	O
36	DUMMY	/	/
37	VDDI	Power supply to the I/O	P
38	VDDI	Power supply to the I/O	P
39	VDDI	Power supply to the I/O	P
40	DUMMY	/	/
41	VSN	Negative input analog power for driver IC	P
42	VSN	Negative input analog power for driver IC	P
43	VSN	Negative input analog power for driver IC	P
44	DUMMY	/	/
45	VSP	Positive input analog power for driver IC	P
46	VSP	Positive input analog power for driver IC	P
47	VSP	Positive input analog power for driver IC	P
48	DUMMY	/	/
49	GND	GND	P
50	C2_M	MIPI input	I
51	B2_M	MIPI input	I
52	A2_M	MIPI input	I
53	GND	GND	P
54	C1_M	MIPI input	I
55	B1_M	MIPI input	I
56	A1_M	MIPI I/O	I
57	GND	GND	P
58	C0_M	MIPI I/O	I
59	B0_M	MIPI input	I
60	A0_M	MIPI input	I
61	GND	GND	P

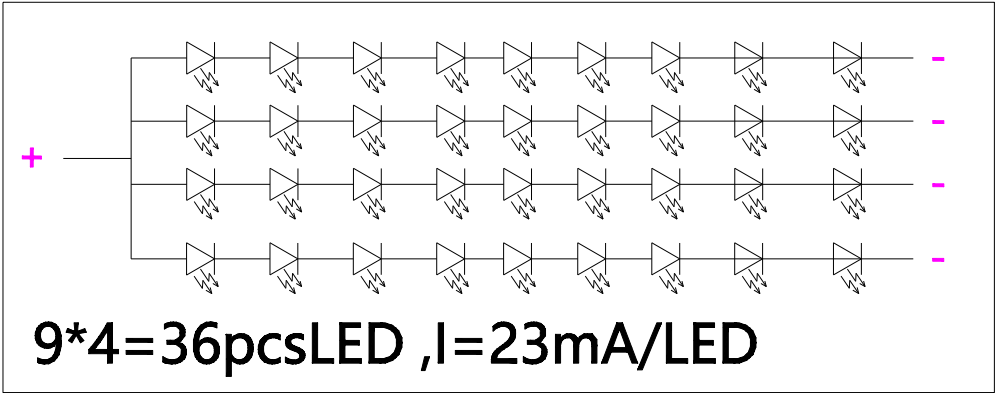
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3.4 Pin assignment for LED Bar

3.4.1 Internal BLU System Connector : soldering

< Table8. Pin assignment for LED Bar >

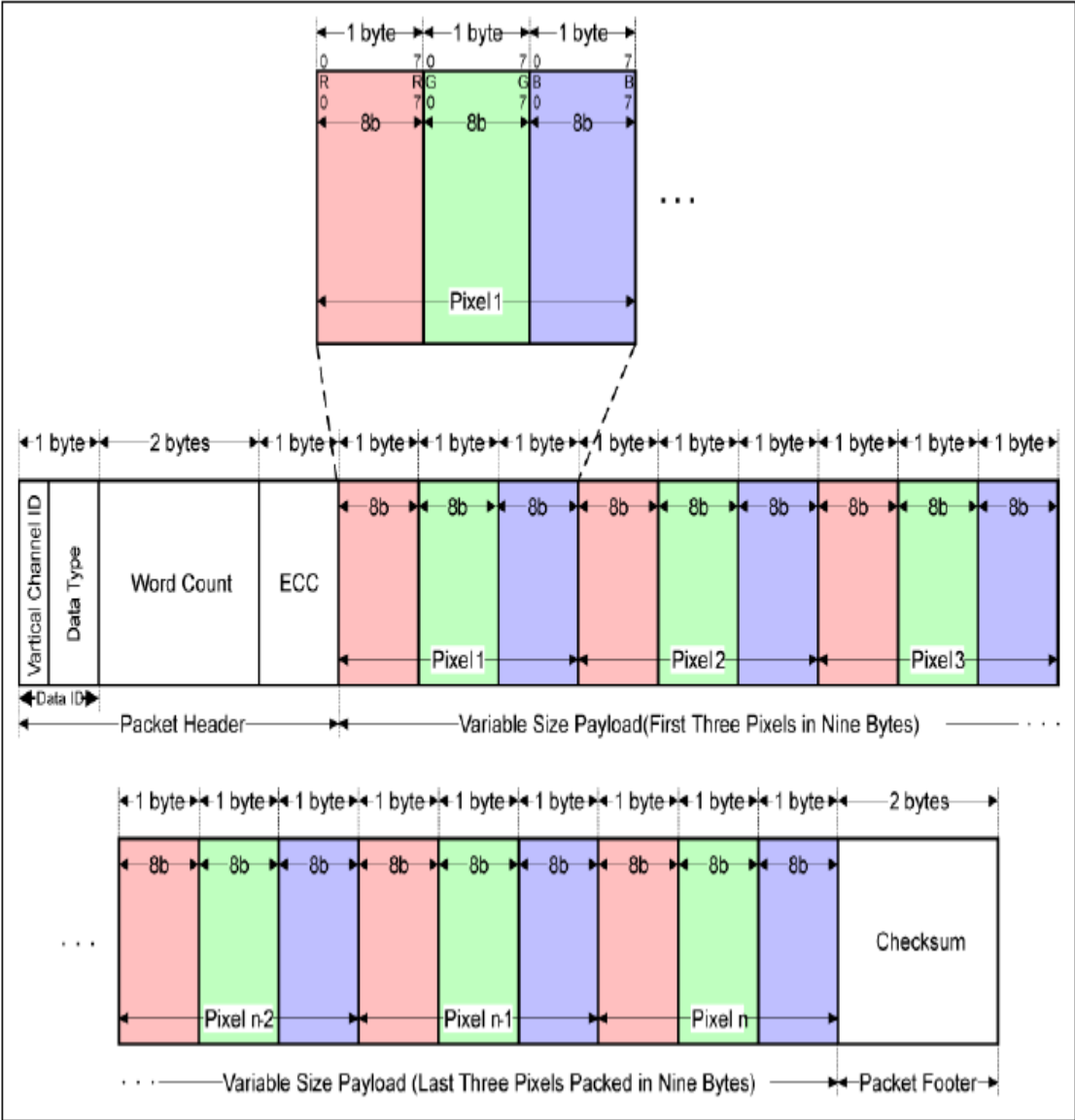
Pin No	Symbol	Description	Remarks
1	FB1	Power for LED1 Cathode	
2	FB2	Power for LED2 Cathode	
3	FB3	Power for LED3 Cathode	
4	FB4	Power for LED4 Cathode	
5	FB5	Power for LED5 GND	
6	FB6	Power for LED6 Anode	



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3.4 MIPI Interface Characteristic

3.4.1 Data Format



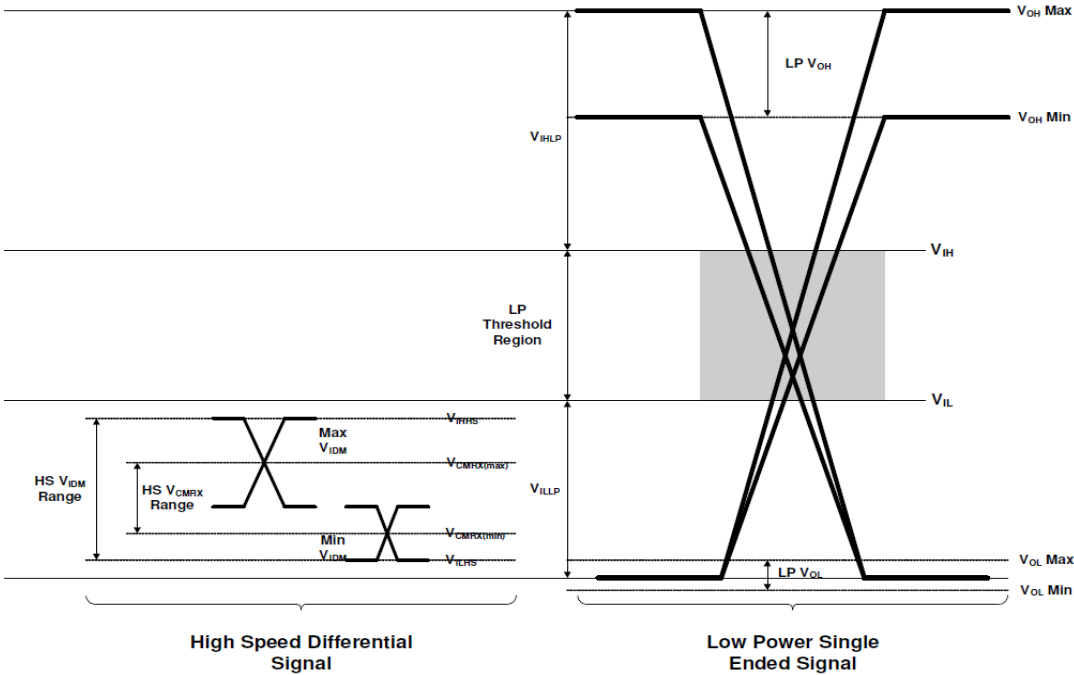
< MIPI Tx Data Configuration >

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3.4.2 DC Specification

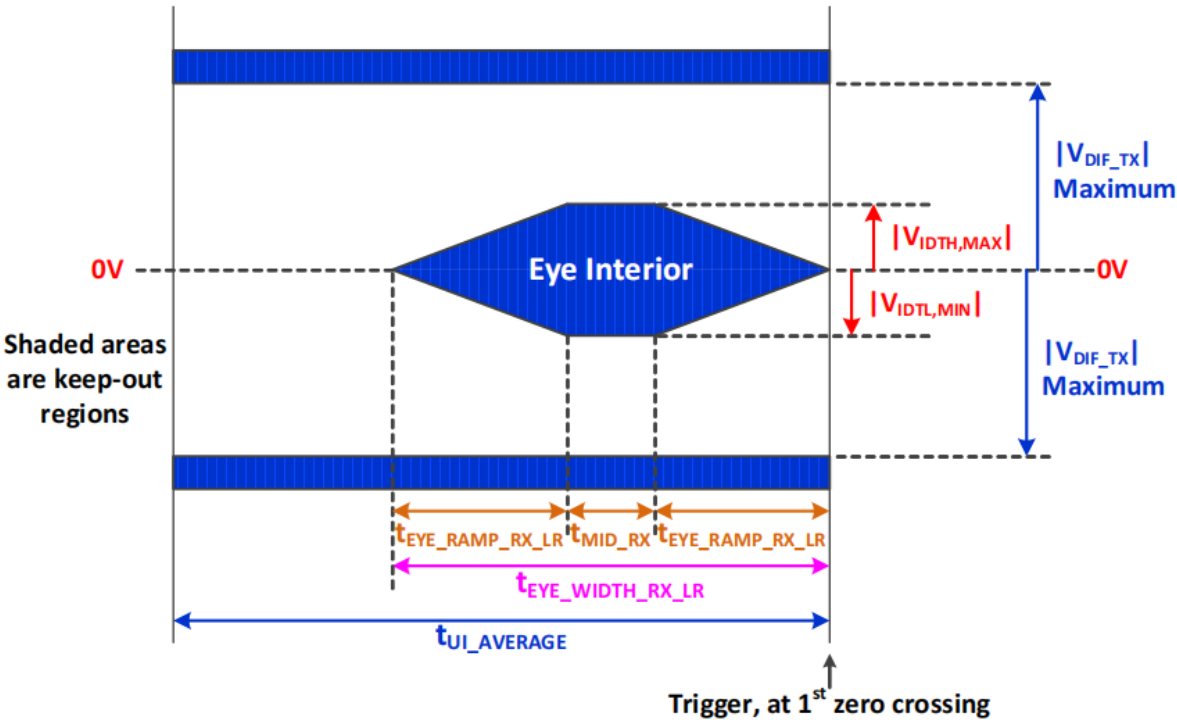
< Table11. DC Specification >

Parameter	Symbol	Min	Typ	Max	Unit	Condition
Power supply voltage for MIPI RX	V _{DDI}	1.65	1.8	1.95	v	
MIPI Characteristics for High Speed Receiver						
Single-ended input low voltage	V _{ILHS}	-40	-	-	mV	
Single-ended input high voltage	V _{IHHS}	-	-	535	mV	
Common-mode voltage	V _{CMRXDC}	95	-	390	mV	
Differential input impedance	Z _{ID}	80	100	120	Ω	
Single-ended threshold for H S termination enable	V _{TERM-EN}	-	-	450	mV	
MIPI Characteristics for Low Power Receiver						
Pad signal voltage range	V _I	-50	-	1350	mV	
Ground shift	V _{GNDSH}	-50	-	50	mV	
Output low level	V _{OL}	-50	-	50	mV	
Output high level	V _{OH}	0.95	1.2	1.3	V	



3.4.3 AC Specification

High Speed Data Transmission: Eye Diagram



Parameter	Symbol	Min	Typ	Max	Units	Notes
UI average	$t_{UI_AVERAGE}$	1		12.5	ns	1,2,5
Eye ramp time at the reference channel output	$t_{EYE_RAMP_RX_LR}$			250	ps	
Eye width at the reference channel output	$t_{EYE_WIDTH_RX}$			Note 3		
Width of the horizontal mid-section of the hexagonal Eye Interior	t_{MID_RX}			Note 4		

Note:

1. This value corresponds to a 1000 Msps data rate.

2. Hexagonal-shaped eye diagram that shall be used at rates below 1Gsps.

3. $t_{EYE_WIDTH_RX_LR} = 2 * t_{EYE_RAMP_RX_LR} + t_{MID_RX} = t_{MID_RX} + 500ps$.

4. $t_{MID_RX} = t_{UI_AVERAGE} - 1000ps$.

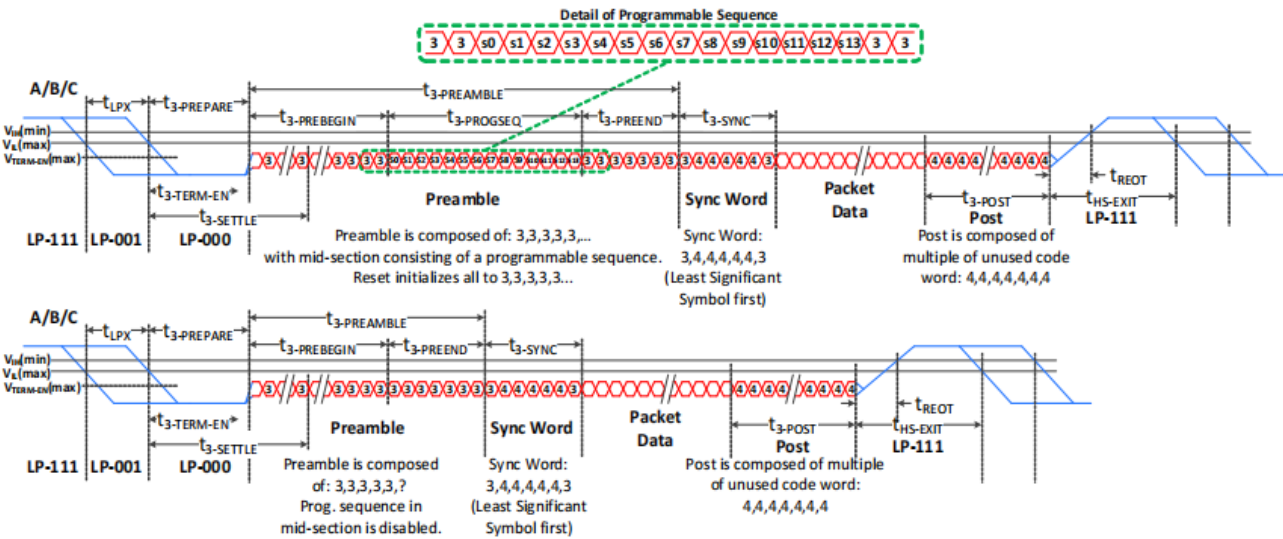
5. For MIPI speed limitation:

[1] Per lane bandwidth is 1G sps

[2] Total Bit Rate: 3Gsps for 8-8-8.

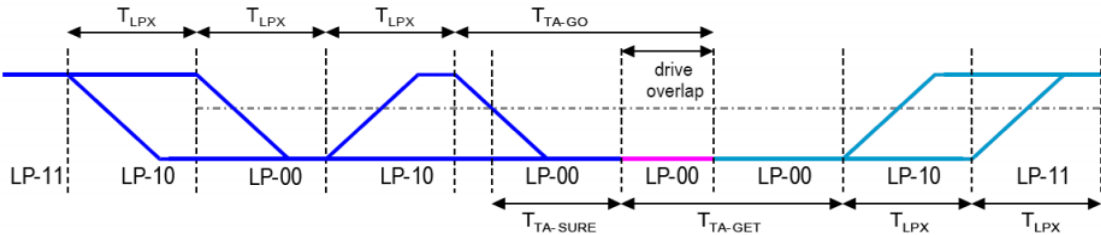
3.5 AC Specification

High-Speed Data Transmission in Bursts



Parameter	Symbol	Min	Typ	Max	Units
Time that the transmitter drives the 3-wire LP-000 line state immediately before the start of the HS transmission.	$t_{3-PREPARE}$	38		95	ns
Time for the slave to enable the HS line termination, starting from the time point when the A, B and C wire cross V_{IL_MAX}	$t_{3-TERM-EN}$	Note4		38	ns
Time interval during which the HS receiver should ignore any HS transitions on the lane, starting from the beginning of $t_{3-PREPARE}$	$t_{3SETTLE}$	95		300	ns
Time that the transmitter drives LP-111 following a HS burst.	$t_{HS-EXIT}$	100			ns
Length of any Low-Power state period	t_{LPX}	50			ns
Time that the transmitter drives 3333333 following LP-000.	$t_{3-PREBEGIN}$		336		UI
Time that the transmitter drives 3333333 before sync word 3444443.	$t_{3PREEND}$	7			UI
Time that the transmitter drives 4444444 after a HSburst	t_{3post}	224			UI

Turnaround Procedure



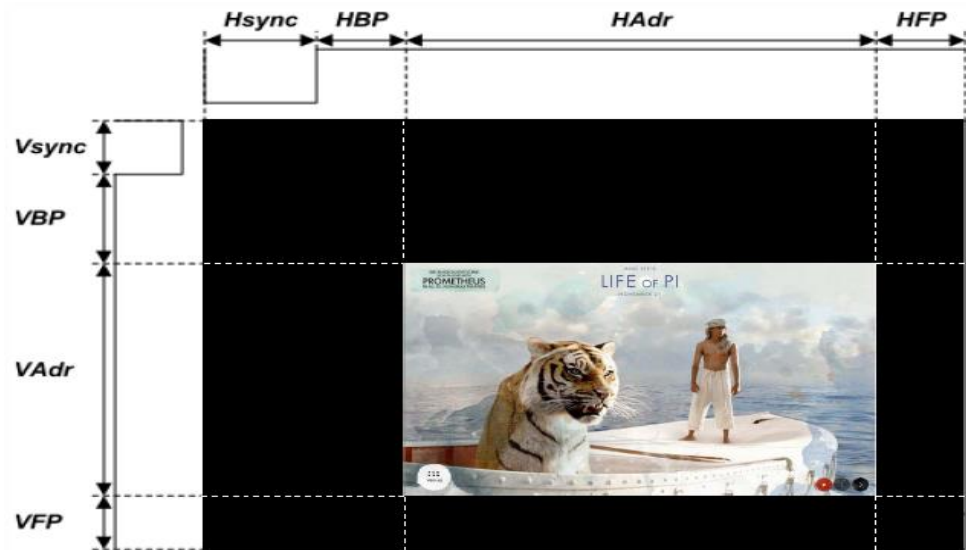
Parameter	Symbol	Min	Typ	Max	Units
Length of any Low-Power state period : Master side	T_{LPX}	50		75	ns
Length of any Low-Power state period : Slave side	T_{LPX}	50		75	ns
Ratio of $T_{LPX}(MASTER)/T_{LPX}(SLAVE)$ between Master and Slave side	Ratio T_{LPX}	2/3		3/2	
Time-out before new TX side start driving	$T_{TA-SURE}$	T_{LPX}		$2T_{LPX}$	ns
Time to drive LP-00 by new TX	T_{TA-GET}		$5T_{LPX}$		ns
Time to drive LP-00 after Turnaround Request	T_{TA-GO}		$4T_{LPX}$		ns

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3.5 Interface timing Parameter

< Table13. Timing Parameter >

Item			Symbol	typ				UNIT
LCD	Frame Rate		-	165/144 /120	90	60	30	Hz
	Pixels Rate		-	589.8				MHz
Timing	DCLK	Frequency	fCLK	853				MHz
		Period	Tclk	1.17				ns
	Horizontal	Horizontal total time	tHP	1702 (DSC)				t _{CLK}
		Horizontal Active time	tHadr	536				t _{CLK}
		Horizontal Pulse Width	tHsync	10				t _{CLK}
		Horizontal Back Porch	tHBP	20				t _{CLK}
		Horizontal Front Porch	tHFP	30				t _{CLK}
		Vertical	Vertical total time	tvp	2850	4560	5700	11400
	Vertical Active time		tVadr	2560				t _H
	Vertical Pulse Width		tVsync	2				t _H
	Vertical Back Porch		tVBP	258				t _H
	Vertical Front Porch		tVFP	30	1740	2880	8580	t _H
Bit Rate			TX SPD (Mbps)	1100				Mbps
Lane				-	2 port 6 trios		-	Lane



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3.6 Input Color Data Mapping

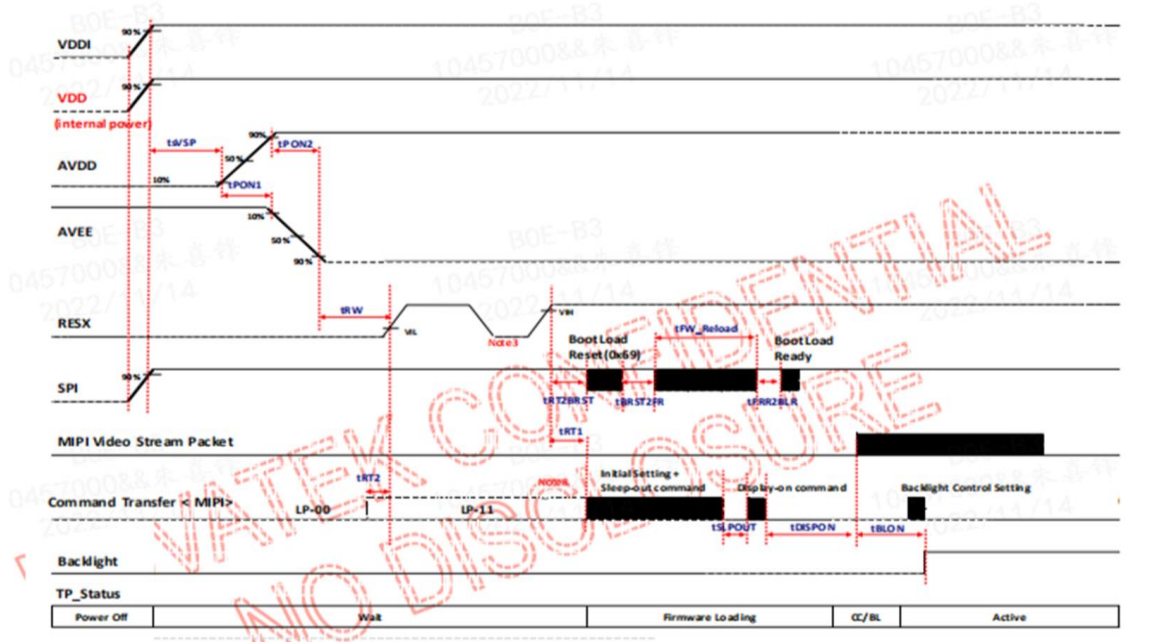
< Table14. Input Signal and Display Color Table >

Color & Gray Scale		Input Data Signal																							
		Red Data								Green Data								Blue Data							
		R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
Basic Colors	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Cyan	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Magenta	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Gray Scale of Red	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	↑								↑								↑							
	▽	↓								↓								↓							
	Brighter	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	▽	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Gray Scale of Green	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	Darker	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
	△	↑								↑								↑							
	▽	↓								↓								↓							
	Brighter	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0
	▽	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
	Green	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Gray Scale of Blue	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	Darker	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
	△	↑								↑								↑							
	▽	↓								↓								↓							
	Brighter	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	1
	▽	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0
	Blue	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Gray Scale of White	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	△	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
	Darker	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0
	△	↑								↑								↑							
	▽	↓								↓								↓							
	Brighter	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1
	▽	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

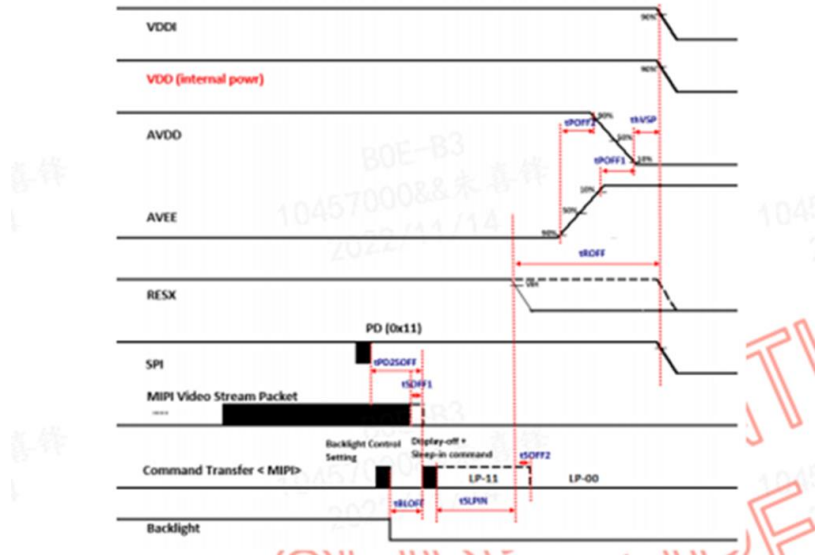
3.7 Power Sequence

To prevent a latch-up or DC operation of the LCD module, the power on/off sequence shall be as shown in below.

Power on sequence



Power off sequence



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3.7 Power Sequence

Power on/off specification

Item	Symbol	Condition	Min	Typ	Max	Unit
AVDD - AVEE delay time (10% to 10%)	tPON1	Power On	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPON2	Power On	0	-	-	us
VDDI On to AVDD On time (90% to 10%)	tsVSP	Power On	1	-	-	ms
All Power On to RESX Hi-level time (90% to VIL)	tRW	Power On	10	-	-	ms
RST Hi-level (VIH) to SPI Boot reset time	tRT2BRST	Power On	10	-	-	ms
Boot load reset command to FW reload	tBRST2FR	Power On	5	-	-	ms
FW reload	tFW_Reload	Power On	65	-	-	ms
FW reload to Boot load ready time	tFRR2BLR	Power On	0	-	-	ms
RST Hi-level (VIH) to 1st Command time	tRT1	Power On	10	-	-	ms
(MIPI goes to LP11 state) to RST goes to Low-level (VIL)	tRT2	Power On	-	0	-	ms
SLPOUT Sequence Request time	tSLPOUT	Power On	100	-	-	ms
MIPI Video Stream On time to BL On time	tBLON	Power On	40	-	-	ms
AVDD - AVEE delay time (10% to 10%)	tPOFF1	Power Off	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPOFF2	Power Off	0	-	-	us
AVDD Off to VDDI Off time (10% to 90%)	tUVSP	Power Off	0	-	-	us
RST Low to VDDI Off time (VIH to 90%)	tROFF	Power Off	0	-	-	us
BL Off to Display Off Command time	tBLOFF	Power Off	0	-	-	us
SLPIN Sequence Request time	tSLPIN	Power Off	60	-	-	ms
VDD regulator turn off time	tVDDOFF	Suspend	80	-	-	ms
MIPI Video Stream Off time to DISPLAY OFF command	tSOFF1	Power Off	0	-	-	us
SPI Power Down to DISPLAY OFF command	tPD2SOFF	Power Off	50	-	-	ms
SPI Wake Up Gesture to DISPLAY OFF command	tWKG2SOFF	Power Off	50	-	-	ms

4.0 OPTICAL SPECIFICATIONS

4.1 Overview

The test of optical specifications shall be measured in a dark room (ambient luminance ≤ 1lux and temperature = 25±2℃) with the equipment of Luminance meter system (CA310 meter system and TOPCON SR-3&RD-80) and test unit shall be located at an approximate distance 50cm from the LCD surface at a viewing angle of θ and Φ equal to 0°. We refer to θ0=0 (=θ3) as the 3 o’ clock direction (the “right”), θ0=90 (= θ12) as the 12 O’ clock direction (“upward”), θ0=180 (= θ9) as the 9 O’ clock direction (“left”) and θ0=270(= θ6) as the 6 O’ clock direction (“bottom”). While scanning θ and/or Φ, the center of the measuring spot on the Display surface shall stay fixed.

4.2 Optical Specifications

< Table16. Optical Table >

Item	Symbol	Condition	Min	Typ.	Max	Unit	Note
luminance-15.6mA	Bp	$\theta=0^{\circ}$	-	500	--	cd/m2	center
luminance-18.8mA	Bp	$\theta=0^{\circ}$	-	600	--	cd/m2	center
luminance-22mA	Bp	$\theta=0^{\circ}$	-	700	--	cd/m2	center
luminance-23mA	Bp	$\theta=0^{\circ}$	640	750	--	cd/m2	center
luminance-25.2mA	Bp	$\theta=0^{\circ}$	-	800	--	cd/m2	center
luminance-29.2mA	Bp	$\theta=0^{\circ}$	-	900	--	cd/m2	center
Brightness Uniformity	ΔBp		70	75	--	%	13P
	ΔBp		70	75	--	%	161P
Viewing Angle	θ_L	$Cr \geq 10$	80	85	--	deg	Single Center Point
	θ_R		80	85	--		
	ψ_T		80	85	--		
	ψ_B		80	85	--		
Contrast Ratio	Cr	$\theta=0^{\circ}$ FF=0°	1000	1340		-	
Response Time	Tr+Tf		--	--	25	ms	Tr+Tf Typ.17ms s Max.20ms GTG Typ.25ms (25°C)
Color Coordinate of CIE1931	Rx	$\theta=0^{\circ}$	0.659	0.679	0.699	-	穗晶2065 KSF 低蓝光
	Ry		0.297	0.317	0.337		
	Gx		0.238	0.258	0.278		
	Gy		0.665	0.685	0.705		
	Bx		0.125	0.145	0.165		
	By		0.042	0.062	0.082		
	Wx		0.279	0.299	0.319		
	Wy		0.2949	0.3149	0.3349		
NTSC Ratio	DCI-P3	CIE1976	92	98	--	%	
	NTSC	CIE1931	96	91	--	%	
Color Temperature	CT		-	7500	-		
Flicker	amount	-	-	-	-30	dB	
Gamma	-		2.0	2.2	2.4		

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Note1:Luminance measurement

- The test condition is at ILED=23*4mA and measured on the surface of LCD module at 25°C.
- The data are measured after LEDs are lighted on for more than 5 minutes and LCM displays are fully white. The brightness is the center value(Point 7) of 5 measured spots (see Figure 8). Measurement equipment CS2000 or similar equipments(Field of view:1deg,Distance:50cm)
 - Measuring surroundings: Dark room.
 - Measuring temperature: Ta=25°C.
 - Adjust operating voltage to get optimum contrast at the center of the display.
 - Measured value at the center point of LCD panel must be after more than 5 minutes while backlight turning on.

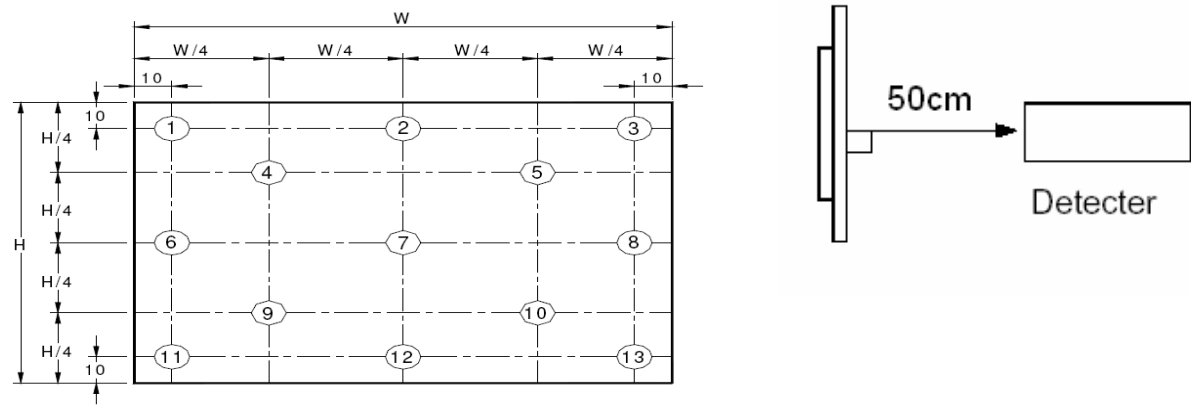


Figure 8. White Luminance and Uniformity Measurement Locations (5 points)

Note2:Uniformity

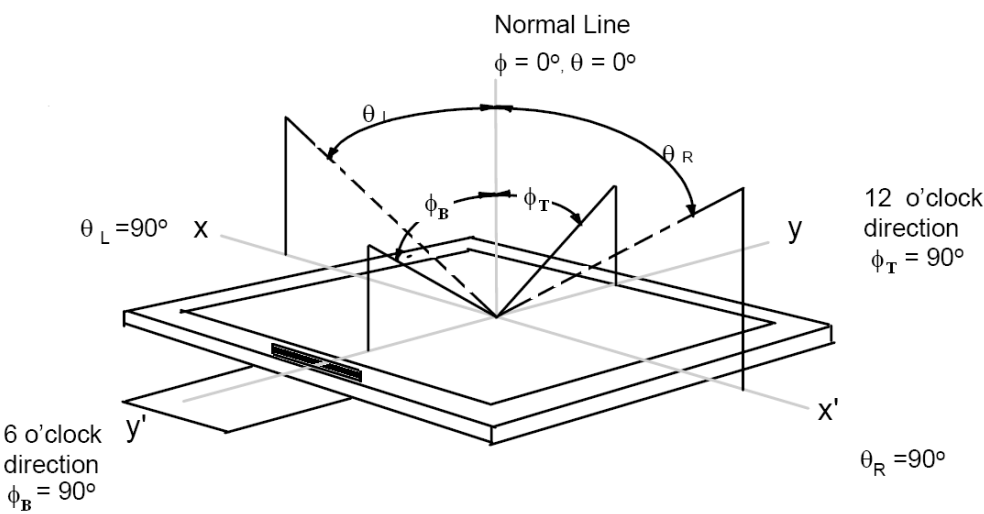
- The test condition is at ILED=23*4mA and measured on the surface of LCD module at 25°C.
- Measurement equipment:CS2000 or similar equipments
- The White luminance uniformity on LCD surface is then expressed as : ΔY_{13} = Minimum Luminance of five points / Maximum Luminance of five points (see Figure 8) , ΔY_{161} = Minimum Luminance of 161 points /Maximum Luminance of 161 points (see Figure 9). (在线添加不了图片, 161P TBD)



Figure 9. Uniformity Measurement Locations (13 points)

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Note 3:The definition of Viewing Angle
Refer to the graph below marked by θ and ϕ .

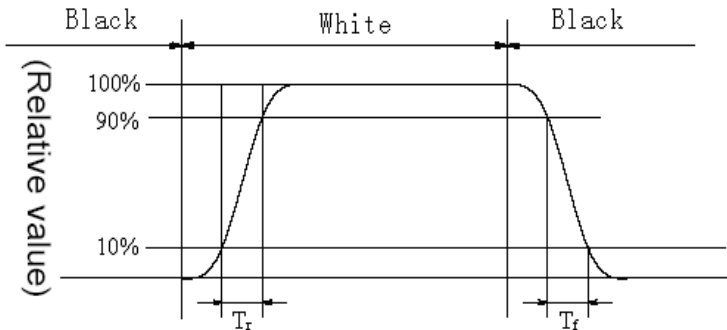


Note4:The definition of Contrast Ratio (Test LCM using CS2000 or similar equipments):

$$\text{Contrast Ratio(CR)} = \frac{\text{Luminance When LCD is at "White" state}}{\text{Luminance When LCD is at "Black" state}}$$

(Contrast Ratio is measured in optimum common electrode voltage)

Note5:Definition of Response time.(Test LCD using DMS501 or similar equipments):
The output sign also photo detector are measured when the input sign also are changed from "black " to "white" (Voltage falling time)and from "white" to "black" (Voltage rising time), respectively . The response time is defined as the time interval between the 10% and 90% of amplitudes . Refer to fi gures below.



	L0	L1	L2	L3	L4	L5	L6	L7
L0								
L1								
L2								
L3								
L4								
L5								
L6								
L7								

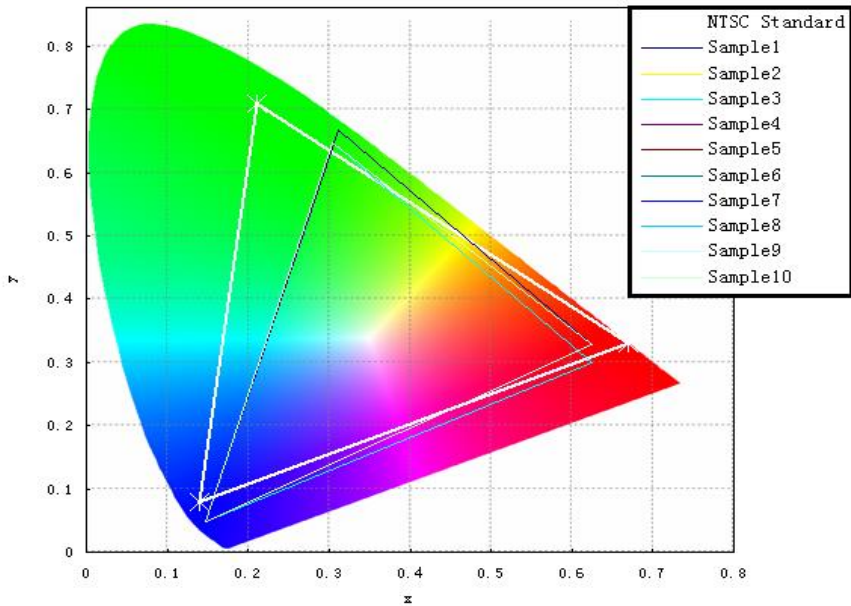
Response time of gray to gray:
Measurement equipment: DMS501 or similar equipments.
Test method: we define 8 grays L0-L7,the grays of L0-L7 were defined as:0,36,73, 109, 146, 182, 219, 25 5. Theoutputsignalsofphotodetectoraremeasuredwhentheinputsignalsarechanged from "Lx" to "Ly" , x, y= [0, 7]. The response time is defined as the time interval between the 10% and 90% of amplitudes. The result of the test can be noted as below:

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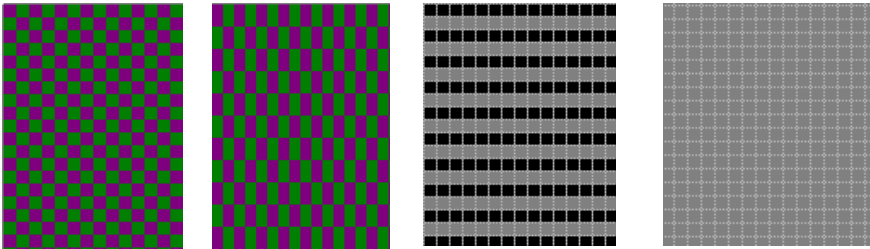
Note 6: Color Coordinates of CIE 1931
 The test condition is at ILED=23*4mA and measured on the surface of LCD module at 25°C.
 Measurement equipment:CS2000 or similar equipments
 The Color Coordinate (CIE 1931) is the measurement of the center of the display shown in below figure.

Note 7: Definition of Color of CIE Coordinate and NTSC Ratio.

$$S=\frac{\text{area of RGB triangle}}{\text{area of NTSC triangle}}\times 100\%$$



Note 8: Flicker
 ●Measurement equipment :CA-310 or similar equipments
 ●Measuring temperature: Ta=25°C.
 ●Test method: JEITA method
 ●Test pattern : Refer to below(Test Pattern should be full-fill of display screen)



1 Dot Inversion, 2 Dot Inversion , Line Inversion , Frame Inversion
 The point should be marked is, for line and frame inversion, the background of Flicker Test Pattern - "gray " are defined as middle gray scale .For example, RGB 24bit "gray" defined as below:

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

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For Dot inversion, the RGB data for first pixel is (127, 0, 127), the RGB data for the second pixel is (0, 127 , 0).

●Frame Frequency Requirement before test: The LCD must be tuned to more than 65HZ before measurement.

●Measurement Point: the center of display active area

●Conversion of Flicker ratio:

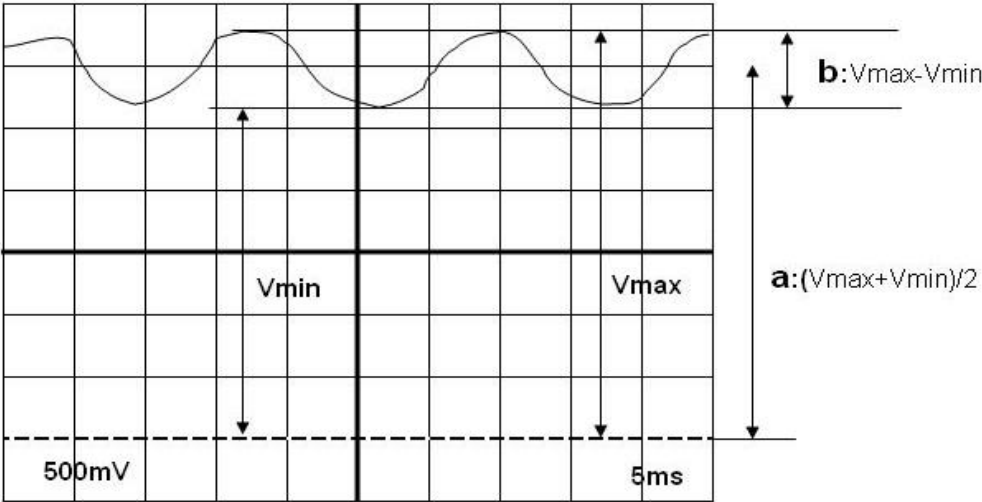
Flicker [dB] = 10 x log[Px/P0]

Where

Px: Maximum power spectrum of AC component after passing through integrator

P0: Power spectrum of DC component after passing through integrator

AC component=b (Refer to below diagram)



Note 9: gamma curve control

- For gamma curve control, Lenovo’ s request as below:
- 1,the whole curve’ s tolerance must control within +/-0.2;

Note 10:Crosstalk

- There should be no visible cross-talk in normal direction of the display when the two " Cross-talk Test Patterns " below are loaded.
- Measurement equipment:CS2000 or similar equipments
- The point should be marked is, the background of Cross-talk Test Pattern- "gray " are defined as middle gray scale . For example, RGB 24bit "gray" defined as below:

R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

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5.0 RELIABILITY TEST

The Reliability test items and its conditions are shown in below.

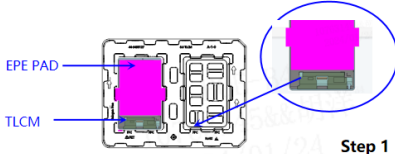
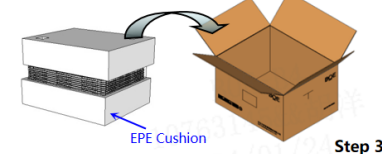
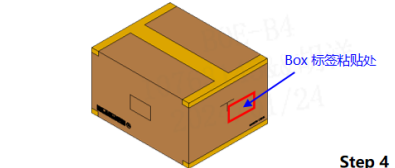
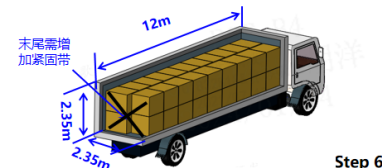
<Table 17. Reliability Test Parameters >

No	Test Items	Conditions
1	高温驱动 (HTO)	60°C, 240hr
2	低温驱动 (LTO)	-20°C, 240hr
3	高温高湿驱动 (HTBO)	60°C, 90%, 240hr
4	高温存储 (HTS)	80°C, 240hr
5	低温存储 (LTS)	-40°C, 240hr
6	高温高湿存储 (HTBS)	60°C, 90%, 240hr
7	残像 (Image Stick)	25°C, 点亮30min, L128, 30s消失

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6.0 PACKING INFORMATION(产品形态：TLCM)

Packing procedure:

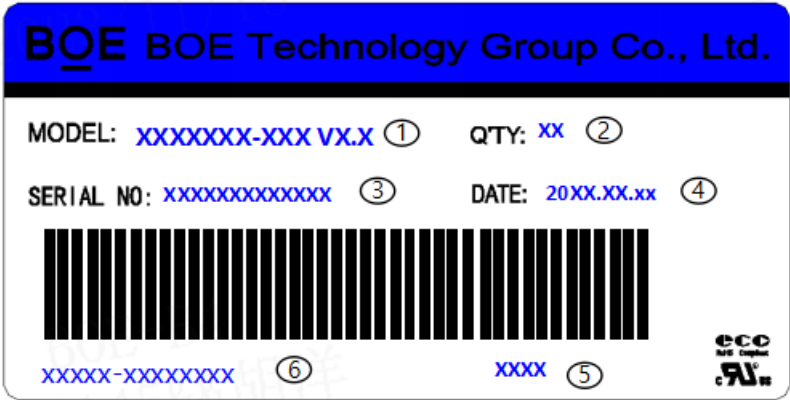
将1ea 2.5t EPE PAD 平放入Tray里，再将一片 TLCM 放在 EPE PAD上面，以此方式依次放入4 ea TLCM (CG朝下放置)，4 ea 2.5t EPE PAD；另一个产品穴如上操作即可。 8 TLCM/10 EPE PAD/Tray (EPE 放置方式如图)	将8pcs Tray直叠堆叠码放，不可旋转，顶部放置 1pcs Tray Cover，并放入PE袋中	将1包Tray平放入Box上下各放置1 pcs EPE Cushion
8 TLCM/10 EPE PAD/Tray (EPE 放置方式如图)	8Tray/ 1Tray Cover/ 1 PE Bag/Box	2 EPE Cushion/Box 64 TLCM/Box
		
采用“H”形封箱方式，对Box进行封箱，并粘贴 Box标签	每个Pallet上放3层Box，每层按照“目目”字形码放6箱，长、宽、高方向安放护角，长、宽、高边捆绑2道打包带，Pallet外用缠绕膜对5个面进行缠绕（至少3层缠膜），并粘贴Pallet标签 18 Box/Pallet, 1152 TLCM /Pallet	厢车装载方式：一横一竖双层码放 厢车装载量_12m: 46080 pcs (40托)
		

6.1 Packing Note

- Box Dimension: 495mm(W) x 395mm(D) x 313mm(H)
- Package Quantity in one Box: 64pcs

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6.2 Box label (产品形态: TLCM)

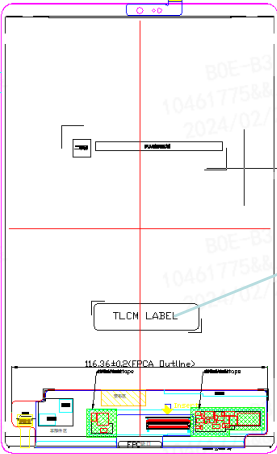


序列号标注部分需打印, 说明如下：

- 1. FG-CODE(前12位)：XXXXXXXX-XXX VX.X
- 2. 产品数量：XX
- 3. Box ID：XXXXXXXXXXXXXXXX
- 4. 包装日期：
- 5. FG-Code后四位：XXXX
- 6. 客户料号：XXXXX-XXXXXXXXX
- 7. Total Size:110×55mm

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7.0 Product Label



5





1

2

3

4

TS088QDM-LL0
XXXXXXXXXXXXXXXXXXXX
8SSD68E26072B1HFYMDXXXX


Label Size: 48mm × 12mm / 厚度: 0.075mm

物料号: 44-9231007

1.FG-CODE : TS088QDM-LL0

2. MDL ID

3. 8S码 : 8SSD68E26072B1HFYMDXXXX

4. MDL ID对应条形码

5. 8S码对应二维码

MDL ID 编码规

序列号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
代码	B	3	A	3	X	X	X	D	9	4	0	X	X	X	X	X	X
描述	GBN代码		等级	B3	年份		月	FG Code后四位				序列号					

8S 编码规则

序列号	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
代码	8	S	S	D	6	8	E	2	6	0	7	2	B	1	H	F	Y	M	D	X	X	X	X
描述	固定值		产品客户端物料号										版本号		供应商产地		年月日			序列号			

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8.0 Handling & Cautions

8.1 Mounting Method

- The panel of the LCD consists of two thin glasses with polarizers which easily get damaged. So extreme care should be taken when handling the LCD.
- Excessive stress or pressure on the glass of the LCD should be avoided. Care must be taken to insure that no torsional or compressive forces are applied to the LCD unit when it is mounted.
- If the customer's set presses the main parts of the LCD, the LCD may show the abnormal display. But this phenomenon does not mean the malfunction of the LCD and should be pressed by the way of mutual agreement.

8.2 Caution of LCD Handling and Cleaning

- Since the LCD is made of glass, do not apply strong mechanical impact or static load onto it. Handling with care since shock, vibration, and careless handling may seriously affect the product. If it falls from a high place or receives a strong shock, the glass may be broken.
- The polarizers on the surface of panel are made from organic substances. Be very careful for chemicals not to touch the polarizers or it leads the polarizers to be deteriorated.
- If the use of a chemical is unavoidable, use soft cloth with solvent (recommended below) to clean the LCD 's surface with wipe lightly.
-IPA(Isopropyl Alcohol), Ethyl Alcohol, Trichlorotrifluoroethane
- Do not wipe the LCD's surface with dry or hard materials that will damage the polarizers and others. Do not use the following solvent.
-Water, Ketone, Aromatics
- It is recommended that the LCD be handled with soft gloves during assembly, etc. The polarizers on the LCD's surface are vulnerable to scratch and thus to be damaged by sharp particles.
- Do not drop water or any chemicals onto the LCD's surface.
- A protective film is supplied on the LCD and should be left in place until the LCD is required for operation.
- The ITO pad area needs special careful caution because it could be easily corroded. Do not contact the ITO pad area with HCFC, Soldering flux, Chlorine, Sulfur, saliva or fingerprint. To prevent the ITO corrosion, customers are recommended that the ITO area would be covered by UV or silicon.

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8.3 Caution Against Static Charge

- The LCD modules use C-MOS LSI drivers, so customers are recommended that any unused input terminal would be connected to Vdd or Vss, do not input any signals before power is turn on, and ground you body, work/assembly area, assembly equipments to protect against static electricity.
- Remove the protective film slowly, keeping the removing direction approximate 30-degree not vertical from panel surface, If possible, under ESD control device like ion blower, and the humidity of working room should be kept over 50%RH to reduce the risk of static charge.
- Avoid the use work clothing made of synthetic fibers. We recommend cotton clothing or other conductivity-treated fibers.
- In handling the LCD, wear non-charged material gloves. And the conducting wrist to the earth and the conducting shoes to the earth are necessary.

8.4 Caution For operation

- It is indispensable to drive the LCD within the specified voltage limit since the higher Voltage than the limit causes the shorter LCD's life. An electro-chemical reaction due to DC causes undesirable deterioration of the LCD so that the use of DC drive should avoid.
- Do not connect or disconnect the LCD to or from the system when power is on.
- Never use the LCD under abnormal conditions of high temperature and high humidity.
- When expose to drastic fluctuation of temperature (hot to cold or cold to hot) ,the LCD may be affected; Specifically, drastic temperature fluctuation from cold to hot ,produces dew on the LCD's surface which may affect the operation of the polarizer and the LCD.
- Response time will be extremely delayed at lower temperature than the operating temperature range and on the other hand at higher temperature LCD may turn black at temperature above its operational range. However those phenomena do not mean malfunction or out of order with the LCD. The LCD will revert to normal operation once the temperature returns to the recommended temperature range for normal operation.
- Do not display the fixed pattern for a long time because it may develop image sticking due to the LCD structure. If the screen is displayed with fixed pattern, use a screen saver.

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8.5 Packaging

- Modules use LCD element, and must be treated as such.
 - Avoid intense shock and falls from a height.
 - To prevent modules from degradation, do not operate or store them exposed directly to sunshine or high temperature/humidity for long periods.

8.6 Storage

- A slight dew depositing on terminals is a cause for electro-chemical reaction resulting in terminal open circuit. Relative humidity of the environment should therefore be kept below 60%RH.
- Original protective film should be used on LCD’ s surface (polarizer). Adhesive type protective film should be avoided, because it may change color and/or properties of the polarizers.
- Do not store the LCD near organic solvents or corrosive gasses.
- Keep the LCD safe from vibration, shock and pressure.
- Black or white air-bubbles may be produced if the LCD is stored for long time in the lower temperature or mechanical shocks are applied onto the LCD.
- In the case of storing for a long period of time for the purpose or replacement use, the following ways are recommended.
 - Store in a polyethylene bag with sealed so as not to enter fresh air outside in it.
 - Store in a dark place where neither exposure to direct sunlight nor light is.
 - Keep temperature in the specified storage temperature range.
 - Store with no touch on polarizer surface by the anything else. If possible, store the LCD in the packaging situation LCD when it was delivered.

8.7 Safety

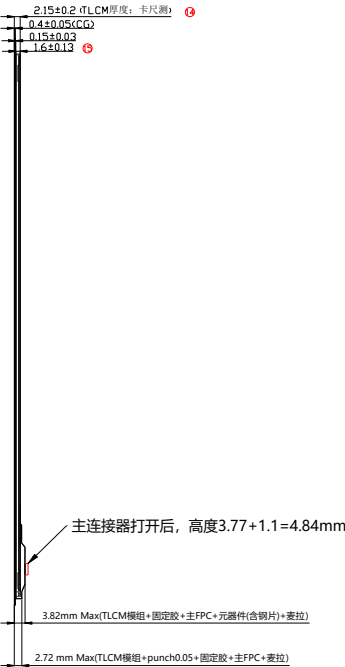
- For the crash damaged or unnecessary LCD, it is recommended to wash off liquid crystal by either of solvents such as acetone and ethanol an should be burned up later.
- In the case the LCD is broken, watch out whether liquid crystal leaks out or not. If your hands touch the liquid crystal, wash your hands cleanly with water an soap as soon as possible.
- If you should swallow the liquid crystal, first, wash your mouth thoroughly with water, then drink a lot of water and induce vomiting, and then, consult a physician.
- If the liquid crystal should get in your eyes, flush your eyes with running water for at least fifteen minutes.
- If the liquid crystal touches your skin or clothes, remove it and wash the affected part of your skin or clothes with soap and running water.

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Mechanical Drawing

Drawing Attachment: Back

侧视图



背视图

